Robin Lange

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SUMMARY

International Mechanical Engineering Honors student from Germany with research experience in autonomous robotic systems, including multi-robot coordination, perception, planning, and data-driven modeling. Developed and tested autonomy software for aerial, wheeled, and legged robots, with experience spanning simulation, sensing, and hardware integration. Interested in advancing robotic autonomy by combining learning-based methods with planning and control and evaluating them on physical systems.

EDUCATION

B.S. Mechanical Engineering, Vehicle Design and Testing Emphasis **GPA: 4.00/4.00**
Georgia Southern University **Aug 2023 – May 2027 · Statesboro, GA**
Honors College Scholar, President's List, Allen E. Paulson Scholar, Tau Beta Pi
Honors Thesis (in progress): Reliability-Gated Lead-Vehicle Prediction for Safety-Supervised Adaptive Cruise Control
Relevant Coursework: Mechatronics, Mechanism Design, Machine Design, Dynamics, Mechanics of Materials, Measurement & Instrumentation Lab, Vehicle Design and Testing, Numerical Methods, Differential Equations, Calculus I–III, Program Design, Circuit Analysis, Logic Circuit Design, FEA, and Engineering Research.

EXPERIENCE

Undergraduate Student Researcher
Intelligent Vehicles Navigation Research Lab, Georgia Southern University **Oct 2023–Present · Statesboro, GA**
- Trained a GRU lead-motion predictor with 27.5% lower held-out forecast RMSE than physics-based extrapolation; integrated causal reliability gating with safety-supervised MPC and investigated switching-induced control transients.
- Built a ROS 2 visual-SLAM/navigation pipeline for indoor UAV inspection using Jetson and RealSense hardware; measured 14.5 cm localization RMSE during a 110-second ground-based trial over valid-reference intervals.
- Developed a LiDAR hand-trajectory interface translating 3D operator movement into UAV guidance commands; demonstrated the prototype and earned second place in the Georgia Power Innovation Awards.

Collaborative Robotics Autonomy Intern
IHMC National Center for Collaborative Autonomy **May – Aug 2026 · Pensacola, FL**
- Built a reusable Rust/Python autonomy framework for heterogeneous robots, connecting Joint Activity Graph decisions, Zenoh state sharing, and capability-based task allocation.
- Resolved motion-capture-to-robot frame inconsistencies and implemented staged Spot grasping, combining mocap-guided positioning with arm-camera refinement for object pickup.
- Developed hybrid simulation and runtime safeguards; validated end-to-end collection-and-delivery missions with two Boston Dynamics Spot robots in more than 10 hardware trials.

Vehicle Testing Research Intern
CARISSMA Institute of Safety in Future Mobility **Jul – Aug 2025 · Ingolstadt, Germany**
- Designed and built a collapsible fold-away truck-rear soft target for underride crash tests, with a magnetically retained, weight-stabilized upper section and two-person assembly.
- Investigated weak radar returns from the original foam target; repositioned tension straps, upper covering, and foam corner elements to improve observed radar response in poorly detected regions.
- Tested LiDAR and 4D-radar responses to covering and reflector arrangements; concealed metal strap fasteners to reduce unwanted reflections and refined the design for more repeatable testing.

Autonomous Systems and AI Intern
Fraunhofer Institute for Transportation and Infrastructure Systems **Jun – Aug 2024 · Ingolstadt, Germany**
- Developed a hybrid Python/ROS 2 UAV mission executive combining explicit finite-state transition guards with reactive behavior-tree actions to simplify adding mission stages.
- Verified transitions among all defined mission states and emergency preemption from any state when a configured trigger occurred in Fraunhofer's flight simulator.
- Benchmarked execution time and code size against pure state-machine and behavior-tree designs; chose the hybrid for clearer extension despite the pure behavior tree's faster execution.

PUBLICATIONS

Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing
Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. *Manuscript in preparation.*

Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing
Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. *SAE Technical Paper 2026-01-0058 (2026).* doi:10.4271/2026-01-0058

Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower
Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. *SSRN Preprint (2026; not peer reviewed).* doi:10.2139/ssrn.7067358

Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications
Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** *SAE Technical Paper 2025-01-8025 (2025).* doi:10.4271/2025-01-8025

RESEARCH PRESENTATIONS

Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications
Southern Regional Honors Council Conference, Montgomery, AL (2026)

3D-Space Hand Trajectory Tracking for Intuitive UAV Control
Southern Regional Honors Council Conference, Chattanooga, TN (2025)

Analyzing the Performance of 3D LiDAR and Camera-Based Object Detection Systems and Their Fusion Using AI for Autonomous Systems
Coauthored research poster, Georgia Southern Research Symposium (2024).

Determining Aircraft Final Approach Altitude and Attitude Using a Machine Learning-Based Sensor Fusion System
Coauthored research poster, Georgia Southern Research Symposium (2024).

AWARDS & HONORS

Gulfstream Innovation, Engineering and Flight Excellence Awards — Third Place, Undergraduate
Georgia Southern University, Allen E. Paulson College of Engineering and Computing Research Symposium · 2026

Honorable Mention, Engineering Division Student Research Video Competition
Council on Undergraduate Research (CUR), Engineering Division · 2026

Georgia Power Innovation Undergraduate Awards — Second Place
Georgia Southern University, Allen E. Paulson College of Engineering and Computing Research Symposium · 2025

INVOLVEMENT

Honors Peer Mentor · Georgia Southern Honors College · August 2026 – Present
Physics Teaching Assistant and Tutor · Georgia Southern Physics Department · January 2025 – Present
Community Leadership and Service · Southern Leaders · August 2023 – Present
First-Year Experience Peer Leader · Georgia Southern First & Second Year Experience · August – December 2024
Programming for Engineers Laboratory Assistant · Georgia Southern Electrical Engineering · January – May 2024

SKILLS

Robotics & Autonomy: ROS 2, Nav2, MAVROS, ArduPilot, Zenoh, Joint Activity Graphs, behavior trees/state machines, multi-robot task allocation, coordinate transforms, inverse kinematics.

Hardware & Engineering: Boston Dynamics Spot, Clearpath Husky, iRobot Create 3, NVIDIA Jetson AGX Orin/Nano, RealSense cameras, Ouster LiDAR, 4D radar, OptiTrack/Motive, motor/encoder integration, LabVIEW, SolidWorks, ANSYS (FEA/CFD).

Perception & Learning: Isaac ROS cuVSLAM, visual SLAM, LiDAR/RGB-D perception, point-cloud processing, sensor fusion, custom YOLO training/evaluation, GRU forecasting, ONNX.

Control & Simulation: PID control, safety-supervised MPC, reliability-gated preview, MATLAB/Simulink, AVL Scenario Simulator, Gazebo, Bevy/Avian3D, CARLA replay, MCAP/Foxglove.

Programming & Development: Rust, Python, MATLAB, C++, Arduino, Protocol Buffers, Git, Linux, CUDA, PyTorch, MATLAB Coder, Docker, CI/CD (GitHub Actions/GitLab).